

PROGRAMMING SKILLS

Languages: C++, Java, Python, SQL, Go, R, JavaScript, Rust

Technologies: CUDA, AWS, Azure, Pytorch, MongoDB, Boost-Python, React, Faiss, TensorFlow, PostgreSQL

EXPERIENCE

Yale University

New Haven, CT

Graduate Research Assistant - Advisor Quanquan C. Liu

August 2024 - Present

- Developed the first distributed LEDP- k -core and triangle counting algorithms under Local Edge Differential Privacy (LEDP), leveraging input-sensitive graph properties (degeneracy, max degree) to tighten utility bounds—achieving $\leq 3x$ error for k -core vs. $131x$ baseline and up to **six orders of magnitude** reduction in triangle counting error on billion-edge graphs—all evaluated in a multi-processor simulation environment with realistic communication overhead. Published in **ACM VLDB 2025**.

University of Rochester Medical Center – Office of Research IT

Rochester, NY

Research Data Engineer II - Biostatistics & Computational Biology: McCall Research Group Jan 2024 - August 2024

- Developing intelligent **storage solutions** for large sequencing data in the MicroRNA project, **optimizing data retrieval** for faster analysis and inference.
- Leading the creation of an **open-source end-to-end software** for microglia image analysis in collaboration with the team, packaging research ideas into accessible and usable software.

Massachusetts Institute of Technology – CSAIL Group

Remote

Graduate Summer Researcher - Advisors : Quanquan C. Liu & Julian Shun

June 2023 - August 2023

- Implement a benchmark suite for **privacy-preserving locally adjustable graph algorithms** in **parallel and distributed settings** with the Parallel Computing Group. Code available on request.

Paris Lodron Universität Salzburg – Database Research Group

Salzburg, Austria

Graduate Summer Researcher - Advisor : Martin Schäler

June 2023 - August 2023

- Developed an alignment algorithm for **dynamic bipartite graph matching**, to uncover intricate language patterns and semantic similarities in biblical texts across languages and historical epochs.
- Collaborated with a team of linguists and researchers to integrate natural language processing and set similarity search algorithms into the BOSS project to improve search accuracy. (Under review for VLDB 2026)

University of Rochester – Computer Science Department

Rochester, NY

Graduate Research Assistant - Advisor : Fatemeh Nargesian

July 2021 - May 2023

- Developed and implemented **KOIOS**, a **novel, exact, efficient, and generic** filter verification system for **top- k** set similarity search using semantic overlap, achieving **5.5x enhanced performance** over current state of the art methods. Published in **IEEE ICDE 2023**.
- Developed an algorithm **fair coreset selection** with **400x speedup** over existing ML-based techniques. Evaluated on MNIST, FashionMNIST, CIFAR10, obtaining **70%** accuracy with only **24%** data.
- Worked on developing **Quok**, an innovative system for **approximate query answering** over Open Knowledge. Addressed challenges related to diverse, noisy, and incomplete data. Published in **HILDA 2023**.

California Institute of Technology – Anima AI Lab

Pasadena, CA

Undergraduate ML Researcher - Advisors : Forough Arabshahi & Animashree Anandkumar

June 2020 - April 2021

- Developed a **novel recursive neural network architecture**, Tree Stack Memory Units (Tree-SMU), to enable **compositional generalization** in the domain of mathematical reasoning.
- Evaluated the generalization of Tree-SMU on four different compositionality tests. We showed that Tree-SMU consistently outperforms the compositional generalization of powerful baselines such as transformers, tree transformers and Tree-LSTMs. (**arXiv Preprint**)

University of Washington – Database Group

Seattle, WA

Undergraduate Research Assistant - Advisors: Brandon Haynes, Batya Kenig & Dan Suciu *April 2019 - December 2020*

- Developed high-performance Python API for **LightDB**, accelerating query speed and enabling access to various video data, including VR and AR videos.
- Optimized Python API mapping to low-level constructs, **reducing device transfer time** and delivering faster query execution for users.
- Implemented **boost-python** framework, expanding query expression usage in LightDB and **increasing user adoption**.
- Created **Maimon**, a pioneering system for discovering approximate **MultiValued Dependencies** and acyclic schemas, employing information theory principles. Optimized Maimon to **minimize file scans** by leveraging information theory for MVD pruning and entropy calculations, comparing performance with in-memory and MySQL databases. Published in **ACM SIGMOD 2020**.

EDUCATION

Yale University

New Haven, CT

Ph.D. Computer Science

August 2024 - Present

Yale University

New Haven, CT

M.Phil. Computer Science

August 2024 - March 2026

University of Rochester

Rochester, NY

M.Sc. Computer Science (3.8/4.0) — Full Scholarship & Research Assistantship

August 2021 - December 2023

University of Washington

Seattle, WA

B.Sc. Mathematics (3.0/4.0)

September 2017 - June 2021

PUBLICATIONS

SeDa: Bridging the Gap between Efficient Syntactic and Precise Semantic Search of Similar Passages in Large Text Corpora: Pranay Mundra, Daniel Kocher, Martin Schäler, Nikolaus Augsten (VLDB 2026)

Practical and Accurate Local Edge Differentially Private Graph Algorithms: Pranay Mundra, Charalampos Papamanthou, Julian Shun, Quanquan C. Liu (VLDB 2025)[Paper]

Approximate Query Answering over Open Data: Mengqi Zhang, Pranay Mundra, Chukwubikem Chikweze, Fatemeh Nargesian, Gerhard Weikum. (HILDA 2023)[Paper]

KOIOS : Top- k Semantic Overlap Set Search: Pranay Mundra, Jianhao Zhang, Fatemeh Nargesian, and Nikolaus Augsten. (IEEE ICDE 2023)[Paper]

Mining approximate acyclic schemes from relations: Batya Kenig, Pranay Mundra, Guna Prasaad, Babak Salimi, and Dan Suciu. (ACM SIGMOD 2020)[Paper]

Compositional Generalization with Tree Stack Memory Units: Forough Arabshahi, Zhichu Lu, Pranay Mundra, Sameer Singh, Animashree Anandkumar. (arXiv Preprint)[Paper]

SERVICE

IEEE Transactions on Dependable and Secure Computing: Reviewer

NeurIPS 2027: Reviewer

ICML 2026: Reviewer

SIAM ALENEX 2026: Artifact Evaluation Program Committee

Workshop on Reliable ML (NEURIPS 2025): Program Committee

AWARDS

Yale Ventures Roberts Innovation Fund: 2025-2026 Awardee

RELEVANT COURSES

Computer Science: Computer Programming I, II; Introduction to Database Systems; Database Systems Internals; Data Structures & Algorithms; Linux Fundamentals; Introduction to Artificial Intelligence; Advanced Algorithms; Analytical Methods in Computer Science; Machine Learning, Parallel & Distributed Systems; Computer Networks; Data Mining; Computational Complexity; End to End Deep Learning; Collaborative Programming & Software Design; Introduction to Cryptography. Building Game Engines; Randomized Algorithms; Scalable and Private Graph Algorithms.

Mathematics: Honors Calculus I, II, III; Real Analysis I, II; Linear Analysis; Probability I, II; Differential Equations, Linear Algebra, Numerical Analysis I, II; Modern Algebra I, II; Combinatorial Theory I, II.

Distributed LEDP Graph Algos: Developed a framework to test and deploy local edge differentially private graph algorithms over a distributed network.

KOIOS: Developed a novel, exact and efficient filter verification system for top-k set similarity search agnostic to similarity measure used.

LightDB: A high-performance DBMS for ingesting, storing, and delivering large-scale 360° and light field VR video content with optimized throughput, bandwidth efficiency, and scalability.

AquaDB & SimpleDB: Implemented a multi-user transactional database server written in Go and Java respectively.

Gene Regulatory System: Optimized the code to leverage GPU parallelism using the CUDA framework for the following paper: McMurray et al. Gene network modeling via TopNet reveals functional dependencies between diverse tumor-critical mediator genes.